Antonina Nazarova

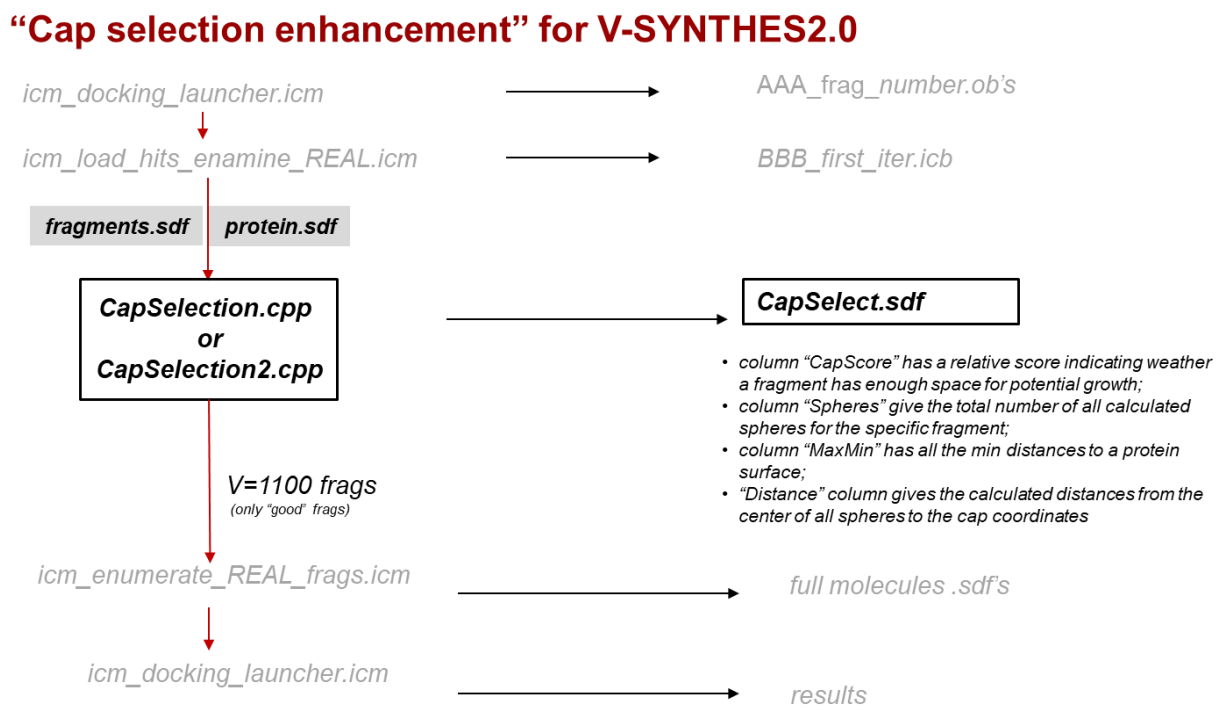
Manual for the fragment selection algorithm “CapSelection” in the V-SYNTHES2.0.

Introduction:

The selection of appropriate fragments having enough space for growth during enumeration into full molecules allows improved enrichment in final docking. The selection algorithm represents an iterative search of the possible vectors of growth which are represented by spheres. Available spheres with certain radii are iteratively searched until either a clash into the wall of the pocket is observed or an exit out of the pocket is detected. The center of the first sphere is defined by the coordinates of the labeled cap of the fragment. The cap defines the point of the growth during enumeration. Apart from the single-labeled caps, multiple labeled (as an aromatic ring) and double-labeled caps are also part of the fragment library.

Improved version of the CapSelection algorithm, CapSelection2 includes consideration of variable vectors of grows when fragment attachment (Figure 1, (B) and (C)).

A)



Executables and example files can be found at /mnt/katritch_lab/Antonina/CapSelection

B) C)

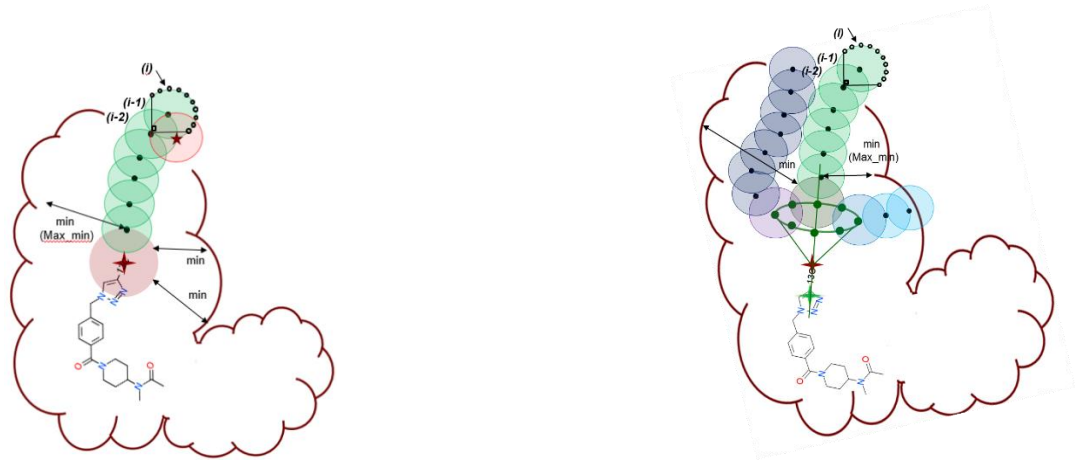


Figure.1. CapSelection enhancement algorithm. A) Schematic representation; B) Spheres-based approach; C) Vector- and spheres- based approach.

Step-by-Step Instructions

Firstly, one needs two .sdf files:

- ***protein.sdf***: containing all the coordinates of interest in the targeted protein structure
- ***fragments.sdf***: containing all the docked fragments

Both of .sdf files are derived from *BBB*_first_iter.icb (the result of running icm_load_hits_enamine_REAL.icm).

The table "*BBB*_first_iter" in the *BBB*_first_iter.icb has to be saved as ***fragments.sdf***. The method for the generation of the ***protein.sdf*** file is described below. Next, execute the executable **CapSelection** or **CapSelection2** (Available both for Windows and Linux (compiled for ubuntu 20.04/18.04)) in the same folder with the ***protein.sdf*** and ***fragments.sdf*** files. The following command should be typed when executing via terminal:

```
nohup ./CapSelection  or nohup ./CapSelection2
```

The program outputs a file called ***CapSelect.sdf***. As the result, the program adds four columns for all entries of the fragments.sdf: "CapScores", "Spheres", "MaxMin" and "Distances". The "CapScores" represents a relative score indicating the potential of the fragment for growth; the "Spheres" column reports the number of possible spheres; the "MaxMin" column contains a list of all minimum distances from a certain sphere to the protein pocket; the "Distance" column gives the calculated distances from the center of all spheres to the original coordinates of the cap.

After opening the ***CapSelect.sdf*** in ICM, the user needs to call the following command in the ICM terminal:

```
add column CapSelect
5.*Log(Abs(CapSelect.Score),2)+Log(Abs(CapSelect.CapScore),2) index=3
name="MergedScore" delete comment="Natural logarithm"
```

The fragments need to be sorted by the newly generated column ”MergedScore in descending order, the first 1100 fragments should be selected for further enumeration. The selected pool of fragments exhibit both a high docking score as well as a high CapScore.

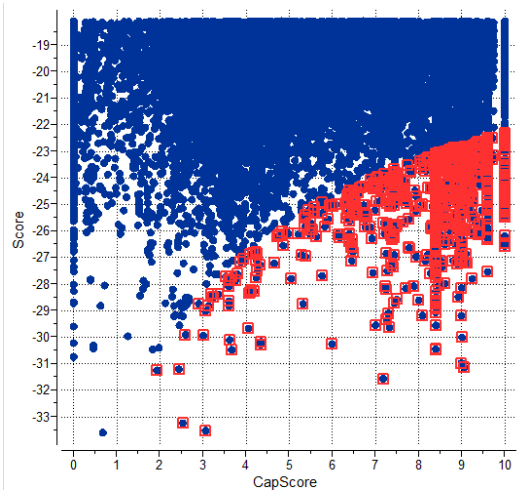


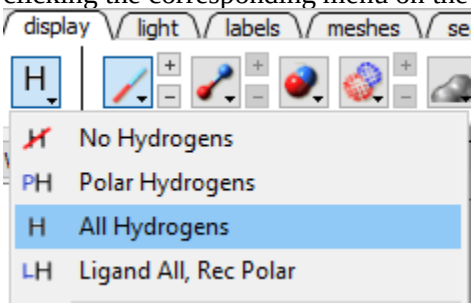
Figure.2. Selection of fragments by MergedScore.

Upon successful execution of the **CapSelection program**, the user should proceed with the fragment enumeration step.

<u>Generation of the **protein.sdf** file.</u>

The user saves those coordinates of the protein that are within the volume of interest defined by an external box around maps (Figure 3, A). Thus, the volume should be slightly larger than the pocket and overall docking maps.

Note: It is crucial to activate the “display hydrogens” option as well as “Toggle Wire Representation” by clicking the corresponding menu on the left top corner of the ICM.



Next, the **protein.sdf** file can be generated by running the following commands in the ICM terminal:

```
String(Chemical( as_graph exact hydrogen  ) mol)
```
will highlight all corresponding atoms

followed by

```
add column t Chemical( as_graph exact hydrogen  )
```
will create table “t”

Lastly, the user needs to save the newly generated “t” table as **protein.sdf** file.
A)

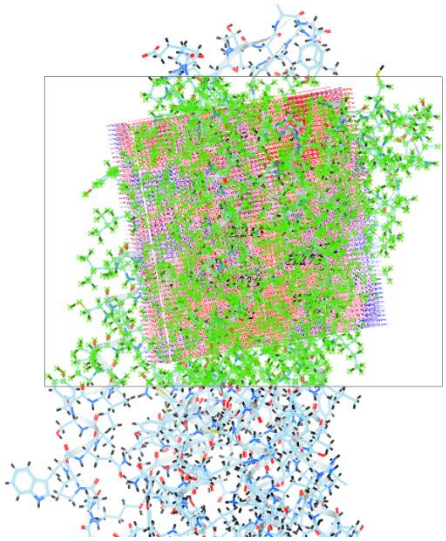


Figure.3. Generation of the **protein.sdf** file. Externally selected protein coordinates are highlighted in green.

When file ***protein.sdf*** has been generated, the user should open it with any text reader app (ex., Notepad++) and verify that hydrogen atoms are present. Specifically, one should pay attention to the availability of hydrogen atoms next to carbon atoms Fig 3, (A). Also, the user needs to specify the number of protein atoms generated in the third line by manually introducing the corresponding number instead of “9999” value (Fig 3, (B)).

A)

```
1   aaaa
2     MOLSOFT AAAAAAAAAAA
3
4   50349999  0  0  1  0  0  0  0  0999 V2000
5      22.4626    2.7452   73.5523 H   0  0  0
6      23.9320    2.4680   75.6920 C   0  0  0
7      23.2658    1.8016   76.2399 H   0  0  0
8      24.2394    1.9399   74.7894 H   0  0  0
9      25.4000    2.7800   76.7070 S   0  0  0
10     26.7200    2.2790   75.5910 C   0  0  0
11     27.1253    1.3204   75.9148 H   0  0  0
12     26.3242    2.1832   74.5799 H   0  0  0
13     20.3210    2.1990   74.9250 O   0  0  0
14     16.9970    2.7350   74.7130 O   0  0  0
15     17.1640    2.2780   71.8690 C   0  0  0
```

'H' must be present next to 'C'

B)

```
1   aaaa
2     MOLSOFT AAAAAAAAAAA
3
4   99999999  0  0  1  0  0  0  0  0999 V2000
5      22.4626    2.7452   73.5523 H   0  0  0
6      23.9320    2.4680   75.6920 C   0  0  0
7      23.2658    1.8016   76.2399 H   0  0  0
8      24.2394    1.9399   74.7894 H   0  0  0
9      25.4000    2.7800   76.7070 S   0  0  0
10     26.7200    2.2790   75.5910 C   0  0  0
11     27.1253    1.3204   75.9148 H   0  0  0
12     26.3242    2.1832   74.5799 H   0  0  0
13     20.3210    2.1990   74.9250 O   0  0  0
14     16.9970    2.7350   74.7130 O   0  0  0
15     17.1640    2.2780   71.8690 C   0  0  0
5034    30.5707   -1.8367   94.2660 O   0  0  0
5035    31.7943   -2.3400   92.4435 N   0  0  0
5036    32.2886   -1.9924   91.6467 H   0  0  0
5037    31.7525   -3.3241   92.6158 H   0  0  0
5038    31.8400    2.2310   90.9970 C   0  0  0
5039   2  3  1
5040   2  4  1
5041   2  5  1
```

Must be changed to N of protein atoms (ex. 5304)

The end of the list of coordinates.
N of protein atoms = 5038 - 5 (lines in the beginning of the file) +1

Fig.4. Generation of the protein.sdf file. A) Presence of carbon connected “H” atoms; B) Change of “9999” or “999” to the selected number of protein atoms (the number can be found by scrolling down through all the coordinates in the .sdf file). The number of the considering protein coordinates should be in the format “AAAA9999”.